

MENOMINEE REGIONAL, MI, USA

WMO: 726487

Lat: 45.117N

Lon: 87.633W

Elev: 625

StdP: 14.37

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 94896

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -8.0	(c) -2.2	(d) -19.7	(e) 1.9	(f) -4.3	(g) -14.8	(h) 2.5	(i) -0.7	(j) 24.1	(k) 24.6	(l) 20.7	(m) 23.2	(n) 5.1	(o) 280	(p) 0.576

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.0	85.8	71.5	82.4	69.3	80.9	67.9	74.0	82.6	71.7	80.3	69.6	77.7	9.9	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
71.1	117.5	80.1	68.8	108.4	77.4	66.3	99.0	75.0	37.9	82.2	35.7	80.4	34.1	78.4	81.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
20.7	18.4	16.5	-13.7	90.8	6.6	3.8	-18.4	93.6	-22.2	95.8	-25.9	97.9	-30.7	100.6
			-13.7	77.0	4.9	2.6	-17.2	78.9	-20.1	80.4	-22.8	81.9	-26.3	83.8
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	44.3	17.8	20.8	29.9	40.9	53.5	63.7	68.8	66.8	59.7	48.0	35.8	24.2
	DBStd	19.75	11.15	11.00	10.75	8.15	8.06	7.30	5.97	5.49	7.73	8.26	9.09	10.26
	HDD50	4171	998	817	629	290	52	1	0	0	10	141	433	800
	HDD65	7930	1463	1237	1089	722	370	110	31	43	194	530	876	1265
	CDD50	2081	0	0	5	18	160	411	581	520	301	78	7	0
	CDD65	365	0	0	0	0	12	70	147	98	35	3	0	0
	CDH74	2782	0	0	2	7	134	611	1180	623	209	16	0	0
	CDH80	657	0	0	0	1	24	162	317	116	36	1	0	0
(14) Wind	WSAvg	7.5	7.5	7.9	8.2	9.2	8.3	7.1	6.5	6.3	7.0	7.6	7.6	7.1
(15) Precipitation	PrecAvg	30.0	1.5	1.0	1.7	2.6	3.5	3.9	3.5	3.0	3.2	3.0	2.0	1.6
	PrecMax	37.7	5.4	2.2	4.0	4.7	5.7	8.8	6.5	5.6	6.2	7.1	5.7	3.5
	PrecMin	23.0	0.4	0.2	0.4	0.7	1.4	1.3	1.1	0.5	1.3	0.4	0.2	0.2
	PrecStd	3.6	1.1	0.5	0.9	1.2	1.2	2.2	1.5	1.3	1.4	1.7	1.3	0.9
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	44.9	48.6	64.5	73.0	82.4	89.4	90.7	87.9	84.3	75.4	63.4	51.0
		MCWB	39.2	42.8	54.1	57.3	65.3	73.7	75.2	72.9	71.9	64.0	52.9	47.6
	2%	DB	39.0	42.9	55.1	64.1	78.3	84.1	86.3	82.4	79.4	70.3	56.5	44.7
		MCWB	34.8	36.9	46.9	51.5	62.5	69.7	72.2	69.8	67.7	61.5	49.8	40.3
	5%	DB	36.2	38.2	49.7	59.2	72.8	81.2	83.3	80.8	75.4	65.1	52.4	40.2
		MCWB	33.5	34.0	42.5	47.7	60.3	67.4	70.8	68.2	65.6	57.5	46.7	36.8
	10%	DB	33.8	35.9	44.9	54.6	68.0	77.2	81.0	77.4	72.6	61.4	48.3	37.0
		MCWB	31.4	32.4	38.5	45.0	57.0	64.9	68.5	66.1	64.1	55.0	43.9	34.1
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	40.4	43.9	57.1	59.3	68.9	75.7	77.7	75.3	73.5	68.5	55.2	48.2
		MCDB	43.9	47.3	65.0	72.0	79.2	85.5	87.9	83.3	81.3	73.2	61.7	50.5
	2%	WB	35.9	37.8	48.4	53.3	65.4	71.6	74.6	72.4	70.1	62.4	51.0	40.8
		MCDB	38.5	41.8	54.7	62.0	74.1	82.3	83.3	80.6	77.1	67.7	56.1	43.6
	5%	WB	33.3	34.5	42.7	49.3	61.6	68.9	72.2	70.2	67.7	58.8	47.2	37.6
		MCDB	35.9	38.1	48.9	58.2	70.7	78.5	80.7	77.6	74.1	64.5	51.1	40.0
	10%	WB	31.0	32.2	38.6	45.8	58.3	66.4	69.9	68.3	65.5	55.3	44.5	34.6
		MCDB	33.3	35.1	44.0	53.6	67.0	75.0	78.5	75.3	71.6	60.5	48.2	37.3
(35) Mean Daily Temperature Range	5% DB	MDBR	14.3	15.3	16.5	17.1	18.7	18.3	19.0	17.9	18.1	16.0	13.3	12.8
		MCDBR	14.5	15.1	21.8	25.9	25.5	21.9	21.5	20.0	20.8	21.0	17.8	14.1
		MCWBR	12.5	11.8	15.0	16.3	14.5	12.2	12.2	11.3	12.6	14.0	13.3	12.3
	5% WB	MCDBR	14.1	14.0	20.7	24.2	23.5	20.1	19.7	17.4	18.6	18.4	16.3	13.2
		MCWBR	12.7	11.5	15.0	16.5	14.7	12.4	12.2	10.7	12.2	13.7	13.1	12.2
(40) Clear-Sky Solar Irradiance	taub		0.260	0.289	0.328	0.358	0.391	0.393	0.384	0.386	0.362	0.328	0.299	0.270
	taud		2.461	2.358	2.369	2.360	2.323	2.352	2.394	2.382	2.443	2.529	2.530	2.491
	Ebn at Noon		278	285	287	286	279	278	279	274	272	269	259	261
	Edh at Noon		22	29	33	37	39	38	36	36	31	25	21	19
(44) All-Sky Solar Radiation	RadAvg		452	771	1151	1448	1758	1899	1953	1658	1283	789	468	345
	RadStd		47	83	107	137	142	146	83	110	96	81	45	35

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (49 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page